

AAE 451B – A7: Avionics and Electronics

Team 01

9/27/26



School of Aeronautics
and Astronautics

Avionics Summary



Pixhawk 6C Flight Controller



PM02 V3 Power Module



M8N GPS 10 Pin



Standard Precision
MS4525DO Sensor
and PT60 Pitot tube



Spektrum DX6e Transmitter



Spektrum DSMX Receiver

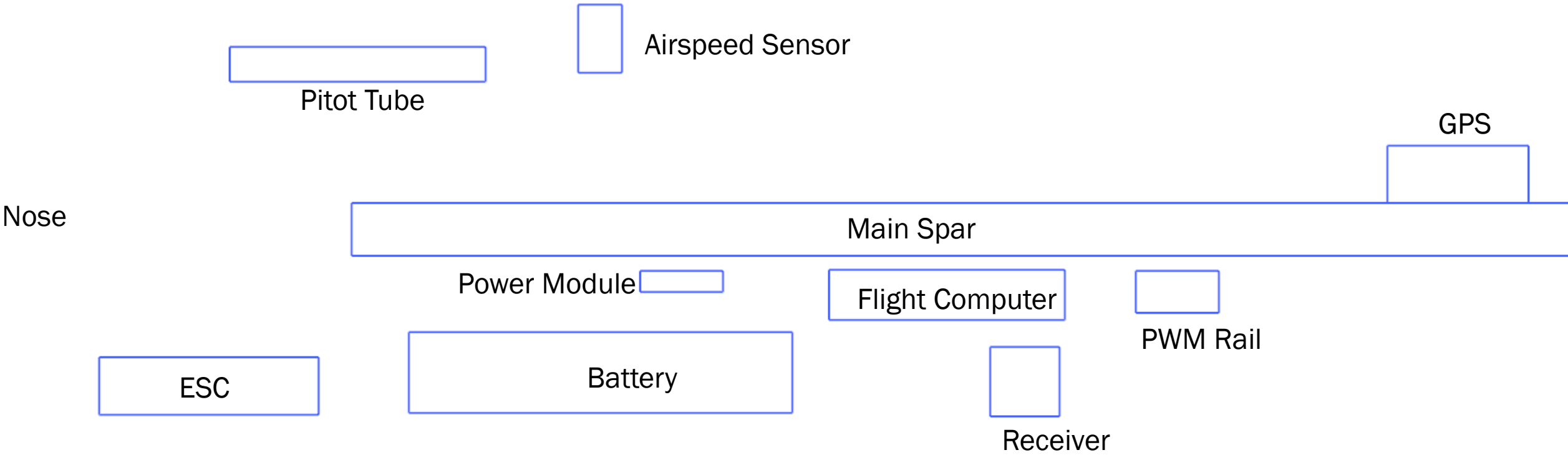


Pixhawk PWM Breakout

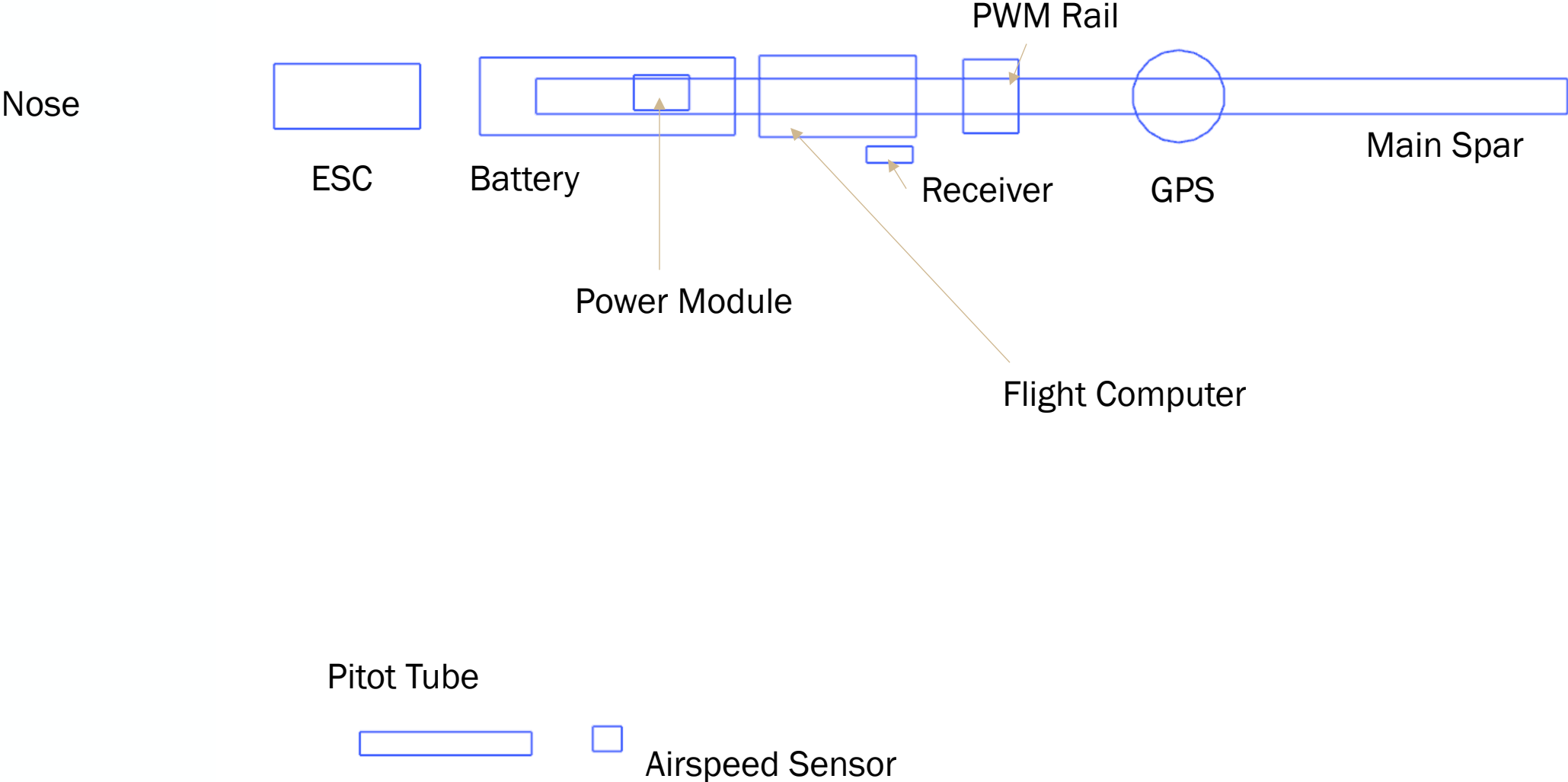
Avionics Summary

Component	Mass (g)	Length (mm)	Width (mm)	Height (mm)	Connector (mm)
Pixhawk 6C	34.6	84.8	44	12.4 (~18 with connectors)	Pixhawk pinned connectors
PM02 V3 Power Module	20	~150	~	~	XT60
M8N GPS	32	50	50	~	10 pin GPS connector
MS4525DO Sensor	4.1	23 (~28 with connector)	13	17	4 Pin I2C
PT60 Pitot tube	15.5	92	27	30	Tubing
Pixhawk PWM Breakout	10	30.1	40	15.5	Servo connectors, BEC connector
Spektrum DSMX Receiver	3.5	25	25	9	DSM 3 Pin

Avionics Layout: Side View



Avionics Layout: Top View



Placement Criteria

Requirement No.	Criteria	Source
1	Pitot tube must be at least 1 foot from fuselage with motor*	From ArduPilot
2	The GPS and receiver must be at least 10 cm from high-current wiring (motor and ESC)	From ArduPilot
3	The GPS must have a clear view of the sky	Required for connection to GPS satellites
4	The flight controller should be within 5 cm of CG	From ArduPilot (“within a few centimeters”)
5	The flight controller should be mounted flat with arrow pointing forward	From ArduPilot
6	The receiver and GPS must be clear of metal structure	Metal structure is not transparent to radio
7	Clearance between each component	Components cannot physically interfere with each other
8	Each item fits in installed location	Components must fit on the vehicle

***Note:** [ArduPilot](#) states that for aircraft that operate with a 25-30% margin to stall speed, pitot tube placement is acceptable not far ahead of leading edge

Placement Check

Requirement No.	Criteria	Pass?
1	Pitot tube must be at least 1 foot from fuselage with motor	Y
2	The GPS and receiver must be at least 10 cm from high-current wiring (motor and ESC)	Y
3	The GPS must have a clear view of the sky	Y
4	The flight controller should be within 5 cm of CG	Y
5	The flight controller should be mounted flat with arrow pointing forward	Y
6	The receiver and GPS must be clear of metal structure	Y
7	Clearance between each component	Y
8	Each item fits in installed location	Y

Avionics Mounting: Overview

- All components except pitot tube and servos mounted to 3D printed trays in fuselage
- Components secured to tray with Velcro straps or zip ties
- Trays glued fuselage panels or bolted to tail boom
- Cables zip-tied to trays as necessary
- Pitot tube bolted to wing spar through 3D printed mount/cover
- Servos mounted to 3D printed inserts glued into cutouts in wings
- PETG used for all 3D printed parts for toughness and strength at higher temperatures

Avionics Mounting: Load Paths

Battery, ESC, controller, receiver, and GPS:



Controller, power module, and PWM breakout:



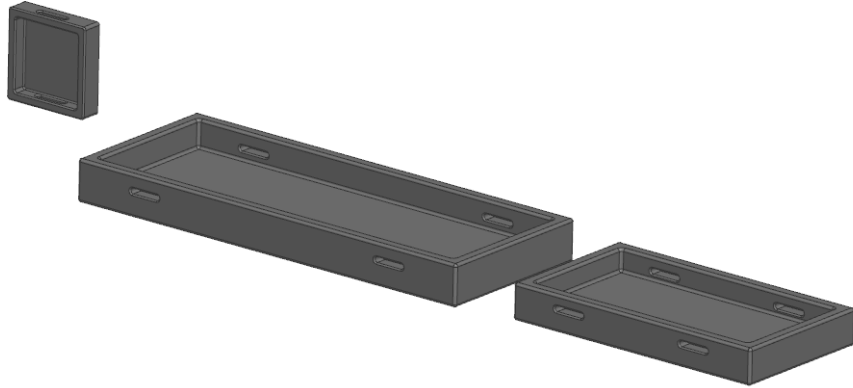
Pitot tube:



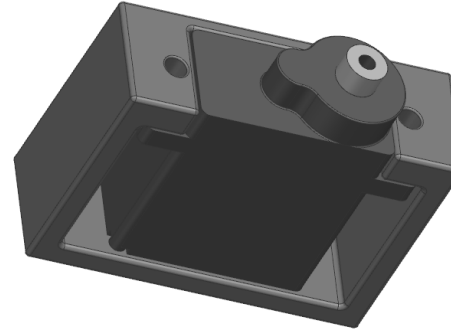
Servos:



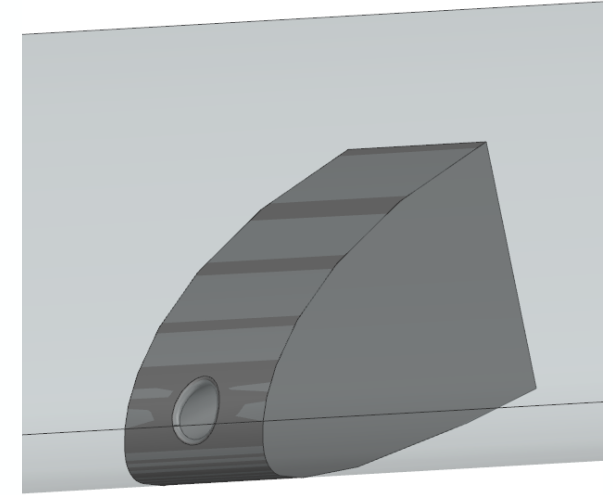
Avionics Mounting Designs



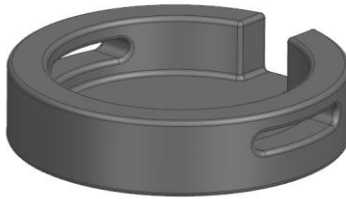
Trays for receiver, battery, and ECS (left to right)



Servo mount



Pitot tube mount



GPS tray



Tray for controller, power module, and PWM breakout

Avionics Mounting Margins: Trays

Limiting case:

- Battery (heaviest component)
- Velcro straps (weakest structure)
- Need at least about 30 mm of Velcro length for positive margins
- Shear strength from Velcro brand
- Load factor of 4 from lectures

Parameter	Value
Component	Battery
Mass	386 g
Load factor	4
Inertial load	15.15 N
Velcro width	10 mm
Velcro contact length	30 mm
Velcro shear strength	0.083 MPa (12 psi)
Factor of safety	1.5
Margin of safety	0.09

Avionics Mounting Margins: Pitot Tube

Limiting case:

- Bearing stress on bolt holes in PETG
- Assumed 1 of 2 bolts carries entire load
- Used tensile strength for margins
 - Bearing strength values for PETG not easily found
 - Tensile strength generally lower than bearing strength
- Load factor of 4 from lectures
- PETG tensile strength from UltiMaker

Parameter	Value
Component	Pitot tube
Mass	15.5 g
Load factor	4
Inertial load	0.61 N
Bolt hole length	4 mm
Bolt diameter	2 mm
Bearing stress	0.076 MPa
PETG tensile strength	40 MPa
Factor of safety	1.5
Margin of safety	349.75

Avionics Mounting Margins: Servos

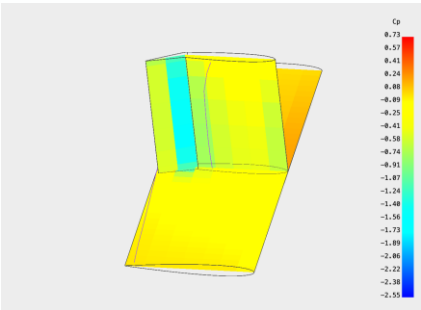
Limiting case:

- Stall torque
- Bearing stress on bolt holes in PETG
- Assumed 1 of 2 bolts carries entire load
- Used tensile strength for margins
 - Bearing strength values for PETG not easily found
 - Tensile strength generally lower than bearing strength
- PETG tensile strength from UltiMaker

Parameter	Value
Component	Servo
Mass	0.2 N-m
Max moment arm	19.4 mm
Load	10.28 N
Bolt hole length	4.2 mm
Bolt diameter	2.5 mm
Bearing stress	0.98 MPa
PETG tensile strength	40 MPa
Factor of safety	1.5
Margin of safety	26.24

Servo Sizing

XFLR & Thin Airfoil Theory Comparison: 35 m/s (Max Speed)



XFLR5 Results

XFLR	Aileron	Elevator	Rudder
Hinge Moment Calculated (Nm)	0.2570	0.2723	0.1969

Thin Airfoil Theory Results

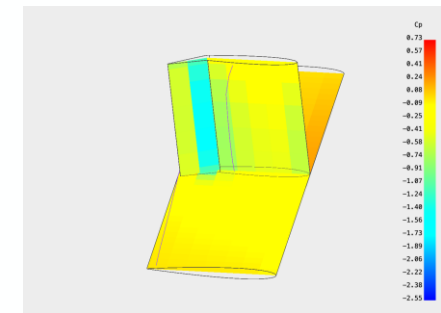
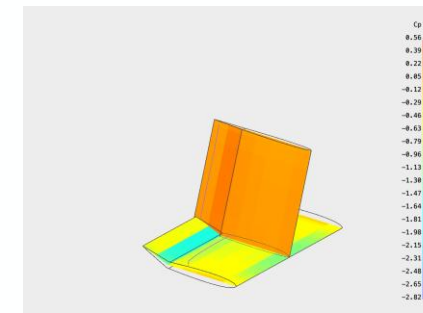
XFLR	Aileron	Elevator	Rudder
Hinge Moment Calculated (Nm)	0.2570	0.4513	0.1836

Thin Airfoil theory is expected to be conservative by a factor of ~2x. We see that it is about ~2x larger than XFLR results for most of the control surfaces. We will design based off XFLR with margin of 1.5.

	Aileron	Elevator	Rudder
Hinge Moment Servo Design (Nm)	0.3855	0.4084	0.2954
... (in-oz)	54.59	57.83	41.83

Cruise Servo Sizing

XFLR & Thin Airfoil Theory Comparison: Cruise 23m/s



XFLR5 Results

XFLR	Aileron	Elevator	Rudder
Hinge Moment Calculated (Nm)	0.063	0.097	.070

Thin Airfoil Theory Results

XFLR	Aileron	Elevator	Rudder
Hinge Moment Calculated (Nm)	0.111	0.210	0.092

Thin Airfoil theory is expected to be conservative by a factor of ~2x. We see that it is about ~2x larger than XFLR results for most of the control surfaces. We will design based off XFLR with margin of 2, or Thin Airfoil (whichever is greater).

	Aileron	Elevator	Rudder
Hinge Moment Servo Design (Nm)	0.146	0.194	0.141
... (in-oz)	20.68	27.47	19.97

Servo Sizing

Servo Selection: 23 m/s – Cruise Speed, (35 m/s – Max Speed)

	Aileron	Elevator	Rudder
Linkage Efficiency (Nylon Clevice)	0.85	0.85	0.85
Linkage Ratio	1:1	1.2:1	1:1
Torque Required (In-ozf) from Servo, Includes Linkage Efficiency Knockdown	24.3 (64.22)	32.3 (68.04)	23.5 (49.2)
Servo Selection: MG90s	2X	1X	1X

Servo Sizing

Servo Specs

Specification	Value
Actuation Limit Stall	180°
Per Unit Weight (g)	13.4
Stall Torque (5.5V) (oz-in)	28.24
Stall Current (mA)	950

MG90S

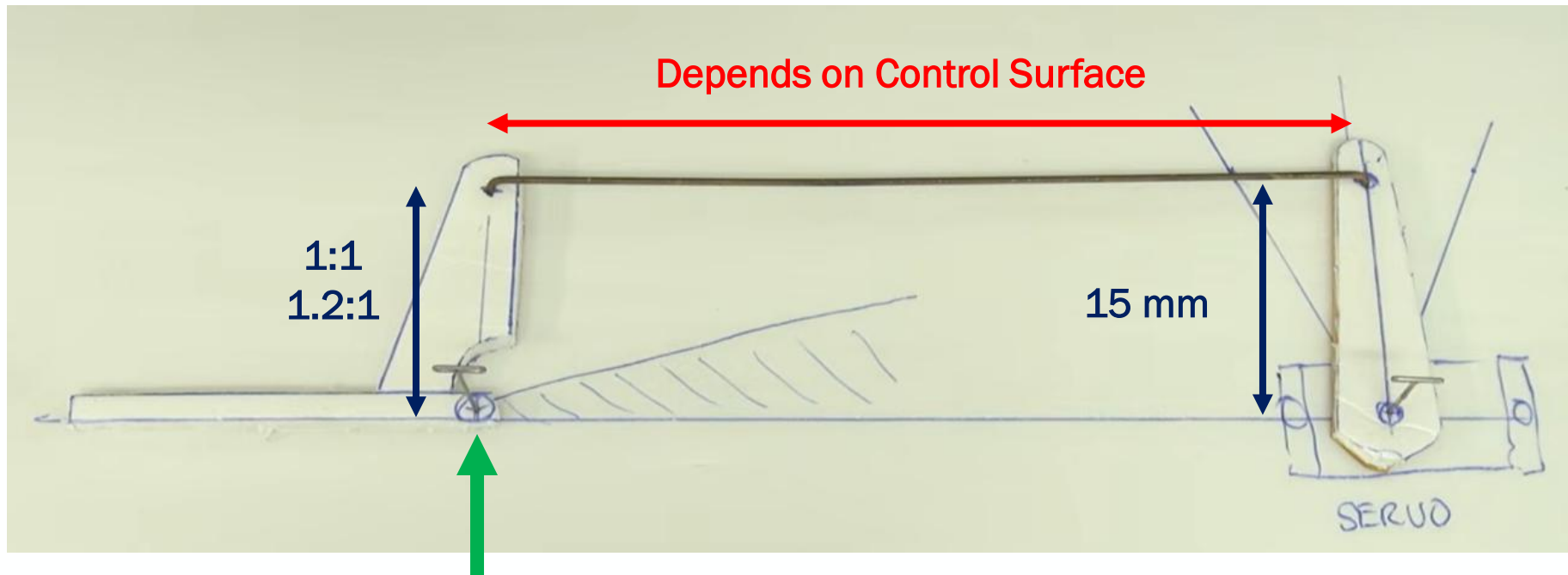


Model

Servo Sizing

Linkage Geometry for Aileron, Rudder, and Elevator

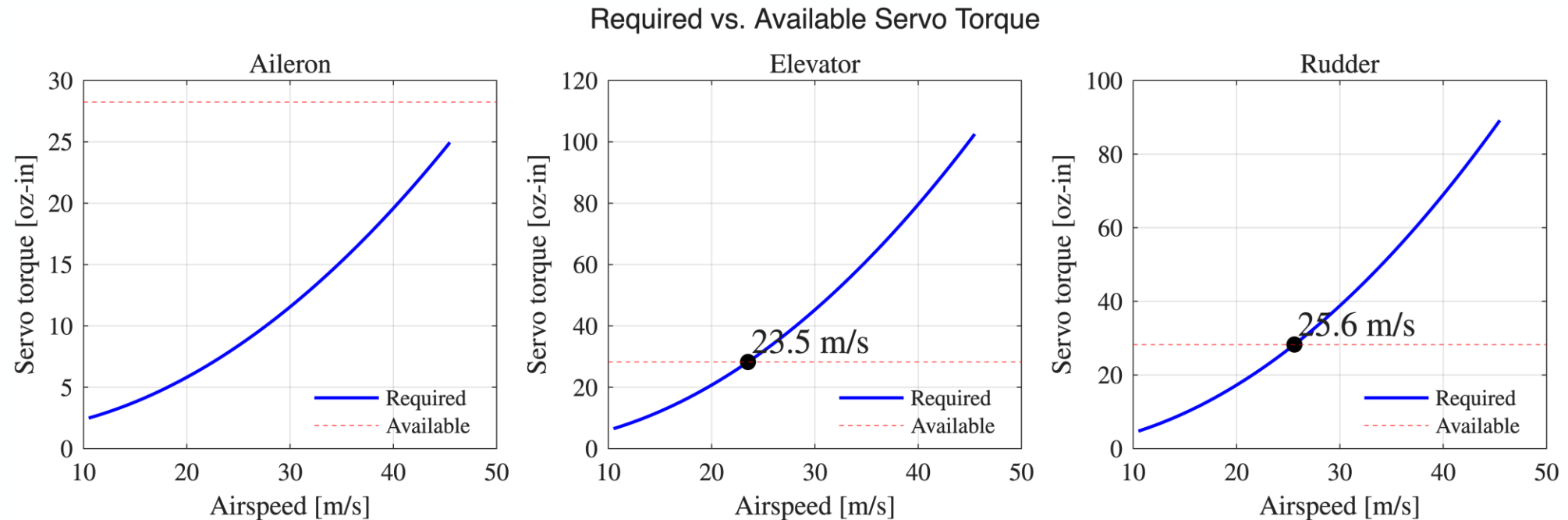
- Servo and Control Horns will have the same length (may increase Control horn length if some additional torque needed)
- Servo and Control Horns will be installed at 90-degree angle per the picture on the below.



Servo Sizing

Available Torques

- Servo available torques are computed using thin airfoil theory. This is known to overestimate by a factor of $\sim 2x$, so additional control authority should exist in real system. Calculated at full deflection.



Servo Sizing

Power Supply

ESC has 5.5V 4A UBEC

Stall torque from Servo = 4.8V < 5.5V supply

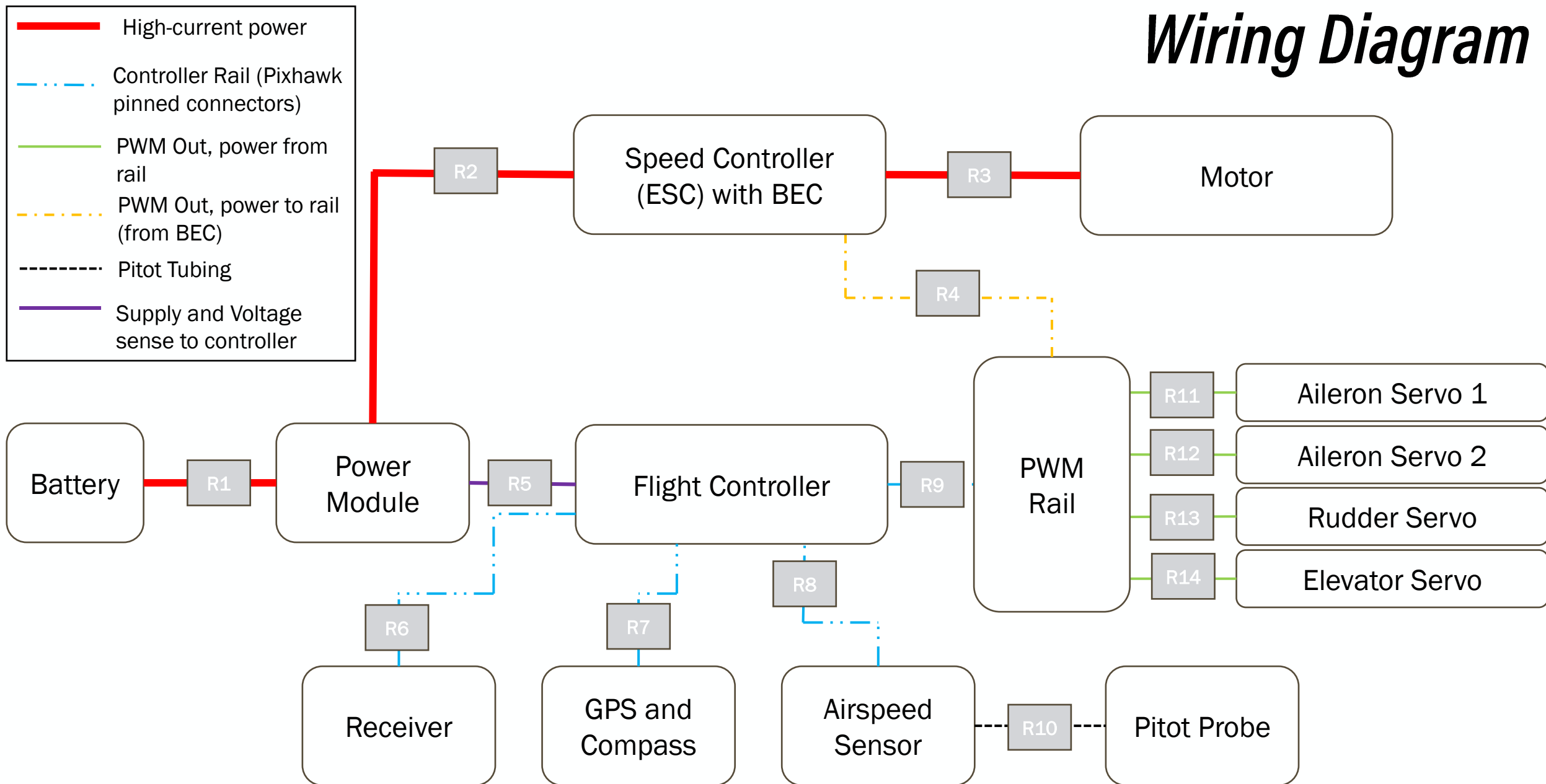
ESC BEC only supplies servo motors, all lines meet voltage requirement

Chosen ESC has a switching power supply

Max power draw = $5.5 * 4 = 22\text{W}$, still under 1000W limit

Max power draw now ~860W

Wiring Diagram



Wiring Schedule

Run	From (port)	To (port)	Gauge	Length (mm)	Mass (g)
R1	Battery Lead	Power Module	12, 2 core	80	5.4
R2	Power Module	ESC	12, 2 core	70	4.7
R3	ESC	Motor	12, 3 core	80	8
R4	ESC PWM	PWM Rail	22, 3 core	230	3.1
R5	Power Module	Flight Computer	28, 6 core	90	2.5
R6	Flight Controller	Receiver	28, 4 core	70	1.2
R7	Flight Controller	GPS	26, 10 core	90	1.9
R8	Flight Controller	Airspeed Sensor	28, 4 core	350	2
R9	Flight Controller	PWM Rail	28, 10 core	80	1.1
R10	Pitot Tube	Airspeed Sensor	Tubing, 2 tubes	50	.5
R11	PWM Rail	Aileron Servo 1	22, 3 core	800	10.8
R12	PWM Rail	Aileron Servo 2	22, 3 core	800	10.8
R13	PWM Rail	Rudder Servo	22, 3 core	650	8.8
R14	PWM Rail	Elevator Servo	22, 3 core	650	8.8

Component Weight Build Up

Component	Mass (g)	Station (+x, +y, +z) measured from wing quarter chord	Source
Avionics			
Flight Controller	34.6	(0, 115.8, -37.7)	Pixhawk
Controller Power Supply	20.0	(-2.6, 16.7, -38.1)	Holybro
GPS	32.0	(0, 150, 0)	Holybro
Airspeed Sensor	4.1	(348.7, -9.2, 54.4)	TE Connectivity
Pitot Tube	15.5	(349.9, -93.1, 45)	Course Supplied
Servo Breakout	10.0	(22.6, 57.1, -57.8)	Pixhawk
Receiver	3.5	(29.2, 144.3, -67.5)	Spektrum
Servo (x2)	13.4	(39.3, 800.4, -14), (0, 800.4, 20)	TowerPro
GPS Tray	4.6	(0, 149.4, 3.0)	3D Printed
Receiver Tray	5.8	(36.1, 142, -68)	3D Printed
ESC Tray	47.3	(0, -149.9, -80.6)	3D Printed
Battery Tray	85.9	(0, 0, -80.3)	3D Printed
Main Tray	66.9	(0, 125, -26.5)	3D Printed
Servo Mount	7.3	(34.4, 800, -11.4)	3D Printed
Pitot Mount	98.8	(349.7, -33.7, 47.1)	3D Printed

Avionics Component Weight Build Up

- Total mass of avionics group: **449.7g**
- Percentage of takeoff mass: **10.24%**
- Change in CG from previous value:
 - +22 mm (towards tail)

Avionics in 3DX

- The model can be found in the 3DX teamwork folder under the name **AAE451 – Team 01 – Configuration 2**

